P02

Decentralized Coordination in Autonomous Swarm Robotics

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Domain DOM_ROB

Abstract

This paper presents algorithms for autonomous multi-robot systems where agents coordinate spatial aggregation and object transportation without direct inter-robot communication. By reading and writing localized environmental markers, the swarm exhibits emergent collective problem solving.

Abstract quoted from M. Dorigo et al., IEEE Transactions on Robotics (2004).

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What is the core problem addressed and what assumptions are made?
2. What mathematical or structural abstractions represent the physical phenomena?
3. How can this concept be mapped or transferred to analogous problems in other domains?